mov esi, A6; size 0x164d
mov ax, [esi+TeamGeneralInfo.resetControls]
or ax, ax

jnz return

mov ax, cpuShotAftertouchTimer
or ax, ax

jz short @@bump_resume_play_ai_timer

FALSE

return

TRUE

sub cpuShotAftertouchTimer, 1

FALSE

mov ax, AI_resumePlayTimer
or ax, ax

jz short @@generate_rand

TRUE

sub AI_resumePlayTimer, 1

FALSE

call Rand
mov ax, word ptr D0
mov AI_rand, ax
mov esi, A6
add [esi+TeamGeneralInfo.AI_timer], 1
mov esi, A6
mov [esi+TeamGeneralInfo.currentAllowedDirection], -1; reset control vars
mov esi, A6
mov [esi+TeamGeneralInfo.firePressed], 0
mov esi, A6
mov [esi+TeamGeneralInfo.fireThisFrame], 0
mov esi, A6
mov [esi+TeamGeneralInfo.quickFire], 0
mov esi, A6
mov [esi+TeamGeneralInfo.normalFire], 0
mov ax, g_inSubstitutesMenu
or ax, ax

jnz return; do only if we're not in subs menu

TRUE

mov word ptr D7, -1
mov esi, A6
mov eax, [esi+TeamGeneralInfo.controlledPlayer]
mov A5, eax; A5 -> controlled player
mov esi, A6
mov eax, [esi+TeamGeneralInfo.passToPlayerPtr]
mov A4, eax; A4 -> player receiving a pass
cmp A5, 0
jz short @@player_direction_set

FALSE

TRUE

mov esi, A5
mov ax, [esi+Sprite.direction]
mov word ptr D7, ax; D7 = controlling player direction

FALSE

mov A0, offset ballSprite
cmp A6, offset bottomTeamData
jz short @@bottom_team

TRUE

mov word ptr D2, 769; lower goal line (for top team)
jmp short @@calc_distance

FALSE

mov word ptr D2, 129; upper goal line (for bottom team)

TRUE

mov word ptr D1, 336; D1 = pitch center by x
mov esi, A0; A0 -> ball sprite
mov ax, word ptr [esi+(Sprite.x+2)]
mov word ptr D3, ax; D3 = ball x
mov esi, A0
mov ax, word ptr [esi+(Sprite.y+2)]
mov word ptr D4, ax; D4 = ball y
mov ax, word ptr D1
sub word ptr D3, ax; D3 = ball x - pitch center x
mov ax, word ptr D2
sub word ptr D4, ax; D4 = ball y - opponents goal line y
mov ax, word ptr D3
mov bx, word ptr D3
imul bx
mov word ptr D3, ax
mov word ptr D3+2, dx; D3 = sqr(ball x - pitch center x)
mov ax, word ptr D4
mov bx, word ptr D4
imul bx
mov word ptr D4, ax
mov word ptr D4+2, dx; D4 = sqr(ball y - opponents goal line y)
mov eax, D4
add D3, eax
mov eax, D3
mov D6, eax; D6 = something of a manhattan distance from the opponent's goal center to the ball
mov esi, A0; A0 -> ball sprite
mov ax, word ptr [esi+(Sprite.x+2)]
mov word ptr D3, ax
mov esi, A0
mov ax, word ptr [esi+(Sprite.y+2)]
mov word ptr D4, ax
mov word ptr D0, 256
push D5
call CalculateDeltaXAndY; calculate angle from the ball toward the middle of the opponents goal
pop D5
jns short @@save_angle

mov word ptr D0, 0; ball right at the center of the goal line

FALSE

mov ax, word ptr D0
mov word ptr D5, ax; D5 = angle between the ball and the center of the opponent's goal
cmp gameStatePl, ST_GAME_IN_PROGRESS
jz @@game_in_progress

TRUE

mov eax, lastTeamPlayedBeforeBreak
cmp A6, eax; is this the team that last played?
jnz return; no, opponents ball, bail

FALSE

mov ax, playingPenalties
or ax, ax

jnz short @@penalty

TRUE

cmp gameState, ST_STARTING_GAME
jb @@game_not_over

FALSE

TRUE

cmp gameState, ST_GAME_ENDED
ja @@game_not_over

FALSE

mov ax, currentGameTick
mov word ptr D0, ax
and word ptr D0, 3Fh
add word ptr D0, 50
add word ptr D0, 50
mov ax, stoppageTimerActive
cmp word ptr D0, ax
ja @@set_direction

TRUE

mov ax, team1Computer; some result is showing
or ax, ax

jz @@game_not_over

FALSE

TRUE

mov ax, team2Computer
or ax, ax

jz @@game_not_over

FALSE

TRUE

cmp gameState, ST_RESULT_ON_HALFTIME
jz short @@showing_result_on_halftime

FALSE

TRUE

cmp gameState, ST_RESULT_AFTER_THE_GAME
jnz short @@not_showing_final_result

FALSE

cmp stoppageTimerTotal, 660
jb return

TRUE

cmp stoppageTimerTotal, 660; showing result after the game for 660 ticks
jb return

FALSE

cmp stoppageTimerTotal, 385; 385 ticks if showing result after 2nd half (but not final)
jb return

TRUE

jmp short @@interval_expired_fire

FALSE TRUE

mov esi, A6; simulate joy fire, to remove the result from the screen
mov [esi+TeamGeneralInfo.firePressed], 1
retn

jmp short @@interval_expired_fire

FALSE TRUE

FALSE

TRUE

jmp short $+2

mov ax, resultTimer
or ax, ax

jnz return

cmp gameState, ST_GOAL_OUT_LEFT
jz @@update_turn_direction

FALSE

TRUE

cmp gameState, ST_GOAL_OUT_RIGHT
jz @@update_turn_direction

FALSE

mov ax, currentGameTick
mov word ptr D0, ax
and word ptr D0, 0Eh; timer mod 16, but only 0,2,4,6,8,10,12,14
jnz return

TRUE

cmp gameState, ST_KEEPER_HOLDS_BALL
jz @@update_turn_direction

FALSE

TRUE cmp gameState, ST_THROW_IN_FORWARD_RIGHT
jb short @@no_throw_in

FALSE

TRUE

cmp gameState, ST_THROW_IN_BACK_LEFT
jbe @@update_turn_direction

FALSE

cmp gameState, ST_FOUL
jz short @@foul_or_free_kick

TRUE

TRUE FALSE

cmp gameState, ST_FREE_KICK_OUTER_LEFT
jb return

FALSE

mov ax, word ptr D5; D5 = angle between the ball and the center of the opponent's goal
mov word ptr D0, ax
add word ptr D0, 16
and word ptr D0, 0FFh
shr word ptr D0, 5; clamp it into 0..7 range, rounding not truncating
mov ax, word ptr D0
mov esi, A6
mov [esi+TeamGeneralInfo.currentAllowedDirection], ax; force only that direction - toward opponents goal
jmp return

TRUE

cmp gameState, ST_FREE_KICK_OUTER_RIGHT
ja return

FALSE

TRUE FALSE

TRUE

TRUE

mov ax, playingPenalties
or ax, ax

jnz @@doing_penalties

FALSE

cmp gameState, ST_PENALTY
jz @@doing_penalties

FALSE

mov ax, AI_rand
mov word ptr D0, ax
and word ptr D0, 7
mov cl, byte ptr D0
mov ax, 1
shl ax, cl; get a random direction mask
test playerTurnFlags, ax
jz short @@penalty_random_direction_disallowed

TRUE

cmp gameState, ST_KEEPER_HOLDS_BALL
jz @@keepers_ball

TRUE FALSE

cmp gameState, ST_GOAL_OUT_LEFT
jz @@keepers_ball

FALSE

mov ax, AI_rand
mov word ptr D0, ax
and word ptr D0, 1; do something random... 2 choices
jz cseg_84775

TRUE

cmp gameState, ST_GOAL_OUT_RIGHT
jz @@keepers_ball

FALSE

TRUE

cmp gameState, ST_PLAYERS_TO_INITIAL_POSITIONS
jz @@goal_scored

TRUE FALSE

cmp gameState, ST_FREE_KICK_OUTER_LEFT
jb short @@test_throw_in

FALSE

cmp stoppageTimerActive, 150
jb cseg_84775

TRUE

cmp gameState, ST_FREE_KICK_OUTER_RIGHT
jbe @@free_kick

FALSE

cmp gameState, ST_THROW_IN_FORWARD_RIGHT
jb short @@test_foul

TRUE

FALSE

mov ax, AI_rand
mov word ptr D0, ax
and word ptr D0, 0Fh
jnz cseg_8470F

TRUE cmp gameState, ST_THROW_IN_BACK_LEFT
ja short @@test_foul

FALSE

cmp gameState, ST_FOUL
jz @@free_kick

TRUE

mov ax, gameState; throw in
mov word ptr D1, ax
sub word ptr D1, ST_THROW_IN_FORWARD_RIGHT
mov A0, offset AI_throwInDirections
mov esi, A0
movzx ebx, word ptr D1
mov al, [esi+ebx]
mov byte ptr D1, al
cmp A6, offset bottomTeamData
jz cseg_84671

TRUE FALSE

ror byte ptr D1, 4; flip direction top-down
jmp cseg_84671

FALSE

mov ax, AI_rand
mov word ptr D0, ax
and word ptr D0, 0Fh
jnz cseg_84775

TRUE

jmp @@apply_after_touch

FALSE TRUE

mov ax, word ptr D7
mov word ptr D2, ax; D2 = controlled player direction
or ax, ax

js return

mov ax, cameraDirection
cmp word ptr D7, ax
jz @@apply_after_touch

FALSE

mov ax, word ptr D7
mov word ptr D0, ax
shl word ptr D0, 5
call FindClosestPlayerToBallFacing
cmp A0, -1; do we have closest player?
jz @@update_turn_direction

TRUE

cmp word ptr ballSprite.x+2, 336
ja short cseg_845CC

FALSE

TRUE cmp word ptr D7, 1
jz @@apply_after_touch

FALSE

cmp word ptr D7, 5
jz @@apply_after_touch

TRUE

cmp word ptr D7, 3
jz @@apply_after_touch

FALSE

TRUE

jmp cseg_84775

TRUE FALSE

cmp word ptr D7, 7
jz @@apply_after_touch

FALSE

TRUE

jmp cseg_84775

FALSE TRUE

mov ax, cameraDirection
cmp word ptr D7, ax
jz @@apply_after_touch

FALSE

TRUE

jmp cseg_84775

TRUE FALSE

mov ax, AI_rand
mov word ptr D0, ax
and word ptr D0, 0Fh
jnz @@apply_after_touch

jmp cseg_84775

TRUE FALSE

mov ax, AI_rand
mov word ptr D0, ax
and word ptr D0, 0Fh
jnz cseg_84775

jmp @@apply_after_touch

FALSE

TRUE

jmp cseg_84775

FALSE

mov ax, word ptr D5
mov word ptr D0, ax
add word ptr D0, 16
and word ptr D0, 0FFh
shr word ptr D0, 5
mov ax, word ptr D7
cmp word ptr D0, ax
jz short @@apply_after_touch

TRUE

mov cx, word ptr D7
mov eax, 1
shl eax, cl
test D1, eax
jnz @@apply_after_touch

FALSE

TRUE

jmp cseg_84775

TRUE FALSE

mov ax, word ptr D0; random direction ok
mov esi, A6
mov [esi+TeamGeneralInfo.currentAllowedDirection], ax
jmp return

FALSE

mov esi, A5
mov ax, [esi+Sprite.direction]
or ax, ax

jz return

TRUE

mov esi, A5
cmp [esi+Sprite.direction], kDirectionBottom
jz return

FALSE

TRUE

jmp short @@apply_after_touch; doing penalties, and the player's direction isn't up or down

FALSE

TRUE

jmp cseg_84815

FALSE

TRUE

mov ax, word ptr D5
mov word ptr D0, ax
add word ptr D0, 16
and word ptr D0, 0FFh
shr word ptr D0, 5
mov ax, word ptr D0
mov esi, A6
mov [esi+TeamGeneralInfo.currentAllowedDirection], ax
retn

shl word ptr D2, 5; D2 = controller player direction, scaled to full byte
mov al, byte ptr D5; D5 = angle between the ball and the opponent's goal center
sub byte ptr D2, al; D2 = difference between player's facing direction and ball->goal center direction
jmp @@update_max_stoppage_time

FALSE

TRUE

mov ax, AI_maxStoppageTime
mov word ptr D0, ax
mov ax, stoppageTimerActive
cmp word ptr D0, ax
ja short @@decide_after_touch_strength

mov esi, A5
mov ax, [esi+Sprite.teamNumber]
mov word ptr D0, ax
mov esi, A0; is this closest player from our team?
mov ax, [esi+Sprite.teamNumber]
cmp word ptr D0, ax
jnz @@update_turn_direction

FALSE

TRUE

jmp @@our_player_closest

FALSE

TRUE

mov esi, A6
mov [esi+TeamGeneralInfo.field_84], 0
mov ax, AI_resumePlayTimer
or ax, ax

jnz return; small delay before getting the ball back to play

mov ax, word ptr D7
mov word ptr D0, ax
shl word ptr D0, 5
call FindClosestPlayerToBallFacing
cmp A0, -1
jz @@apply_after_touch

TRUE

mov esi, A5
mov ax, [esi+Sprite.teamNumber]
mov word ptr D0, ax
mov esi, A0
mov ax, [esi+Sprite.teamNumber]
cmp word ptr D0, ax
jnz @@apply_after_touch

FALSE

TRUE

jmp @@our_player_closest

FALSE

mov ax, AI_turnDirection; pass only if timer mod 16 == 1 or 0
or ax, ax

jns short cseg_8489B

FALSE

TRUE cmp AI_turnDirection, -1
jz short cseg_84890

FALSE

cmp AI_turnDirection, 1
jz short cseg_848BB

TRUE

add AI_turnDirection, 1
cmp AI_turnDirection, -1
jnz return

FALSE

mov word ptr D1, 1
jmp short @@apply_turn_direction

TRUE

FALSE TRUE

mov ax, word ptr D7
mov word ptr D0, ax
mov ax, AI_turnDirection
add word ptr D0, ax
and word ptr D0, 7
mov cl, byte ptr D0
mov ax, 1
shl ax, cl
test playerTurnFlags, ax
jz short @@save_for_next_frame

sub AI_turnDirection, 1
cmp AI_turnDirection, 1
jnz return

FALSE

mov word ptr D1, -1

TRUE

FALSE TRUE

mov ax, word ptr D0
mov esi, A6; seems to turn the player to that direction
mov [esi+TeamGeneralInfo.currentAllowedDirection], ax
retn

FALSE

mov ax, word ptr D1
mov AI_turnDirection, ax
or ax, ax
retn

TRUE

mov ax, penalty
or ax, ax

jz short @@no_penalty

FALSE

mov esi, A6
mov ax, [esi+TeamGeneralInfo.spinTimer]
or ax, ax

jns @@ball_after_touch_allowed

TRUE

FALSE

mov ax, cpuShotAftertouchTimer
or ax, ax

jz short @@direction_updated

TRUE

FALSE

mov esi, A6; possibly apply ball after-touch
mov ax, [esi+TeamGeneralInfo.controlledPlDirection]
mov word ptr D1, ax
mov ax, playingPenalties
or ax, ax

jnz short @@no_ball_after_touch

TRUE

call SetCpuPlayerDirectionTowardGoal

FALSE

cmp A5, 0; check for controlled player
jz return

TRUE

mov esi, A6
mov ax, [esi+TeamGeneralInfo.spinTimer]
or ax, ax

jns @@ball_after_touch_allowed

FALSE

TRUE

mov esi, A6
mov al, [esi+TeamGeneralInfo.plVeryCloseToBall]
or al, al
jnz short @@theres_a_player_near

TRUE FALSE

mov esi, A6
mov al, [esi+TeamGeneralInfo.plCloseToBall]
or al, al
jnz short @@theres_a_player_near

FALSE

call DecideWhetherCpuPlayerFires
jz return

TRUE

jmp @@noone_near

TRUE FALSE

call DecideWhetherCpuPlayerFires
jz return

mov eax, lastTeamPlayed
cmp A6, eax
jz @@noone_near

mov esi, A6
mov eax, [esi+TeamGeneralInfo.opponentsTeam]
mov A0, eax
mov esi, A0
mov ax, [esi+TeamGeneralInfo.playerHasBall]
or ax, ax

jz @@noone_near

FALSE

TRUE

jmp cseg_855B1

FALSE TRUE

jmp cseg_84DD3

mov ax, deadVarAlways0
or ax, ax

jz short cseg_84A0D

FALSE

TRUE

mov esi, A6
mov eax, [esi+TeamGeneralInfo.opponentsTeam]
mov A0, eax
mov eax, deadVarAlways0_1
cmp A0, eax
jnz short cseg_84A0D

FALSE

mov esi, A5
cmp [esi+Sprite.direction], 2
jz short @@player_facing_left_or_right

TRUE

mov esi, A0
mov ax, [esi+TeamGeneralInfo.playerHasBall]
or ax, ax

jnz return

FALSE

TRUE

FALSE

TRUE

mov esi, A5
cmp [esi+Sprite.direction], 6
jnz short cseg_84A53

FALSE

cmp A6, offset topTeamData
jz short @@top_team_test

TRUE

FALSE

cmp D6, 28800
ja cseg_84AEB

TRUE cmp word ptr ballSprite.y+2, 158
jle cseg_84AEB

FALSE

cmp word ptr ballSprite.y+2, 740
jge cseg_84AEB

TRUE

jmp short cseg_84A53

FALSE

mov esi, A6
mov ax, [esi+TeamGeneralInfo.field_84]
or ax, ax

jz short cseg_84B5B

TRUE

FALSE

TRUE

cmp D6, 12800
jb short cseg_84A85

FALSE

TRUE

mov ax, AI_rand
mov word ptr D0, ax
and word ptr D0, 3
jnz short cseg_84AEB

FALSE

mov byte ptr D1, 0Fh
cmp D6, 3200
ja short cseg_84A9F

TRUE

FALSE

TRUE

mov byte ptr D1, 32h

FALSE

mov esi, A5
mov ax, [esi+42]
mov word ptr D2, ax
or ax, ax

js short cseg_84AEB

TRUE

shl word ptr D2, 5
mov al, byte ptr D5
sub byte ptr D2, al
mov al, byte ptr D1
cmp byte ptr D2, al
jg short cseg_84AEB

FALSE

TRUE

neg byte ptr D1
mov al, byte ptr D1
cmp byte ptr D2, al
jg cseg_850F9

FALSE

TRUE

FALSE

mov ax, AI_resumePlayTimer
or ax, ax

jnz return

TRUE

mov esi, A6
sub [esi+TeamGeneralInfo.field_84], 1
mov ax, word ptr D7
mov word ptr D0, ax
shl word ptr D0, 5
call FindClosestPlayerToBallFacing
cmp A0, -1
jz cseg_84D57

FALSE

cmp D6, 9800
jb cseg_84DD3

TRUE

mov esi, A5
mov ax, [esi+Sprite.teamNumber]
mov word ptr D0, ax
mov esi, A0
mov ax, [esi]
cmp word ptr D0, ax
jz @@our_player_closest

FALSE

mov esi, A6
mov al, [esi+TeamGeneralInfo.plVeryCloseToBall]
or al, al
jz cseg_84E16

TRUE

jmp cseg_84D57

TRUE FALSE

cmp A5, 0
jz cseg_84DD3

FALSE

mov esi, A6
mov al, [esi+TeamGeneralInfo.plVeryCloseToBall]
or al, al
jz short cseg_84E16

TRUE

mov esi, A6
mov eax, [esi+TeamGeneralInfo.opponentsTeam]
mov A0, eax
mov esi, A0
mov eax, [esi+TeamGeneralInfo.controlledPlayer]
mov A2, eax
cmp A2, 0
jz short cseg_84BBE

FALSE

TRUE

mov esi, A2
cmp dword ptr [esi+74], 800
jb short cseg_84C00

FALSE

mov esi, A0
mov eax, [esi+TeamGeneralInfo.passToPlayerPtr]
mov A2, eax
cmp A2, 0
jz cseg_84DD3

TRUE

mov esi, A2
cmp dword ptr [esi+74], 5000
jb cseg_84C93

FALSE

cmp D6, 180000
ja cseg_84F4B

TRUE

FALSE

cmp AI_rand, 8
ja short cseg_84CAD

TRUE

mov esi, A2
cmp [esi+Sprite.ballDistance], 800
jb short cseg_84C00

FALSE

TRUE

mov esi, A2
cmp [esi+Sprite.ballDistance], 5000
jb cseg_84C93

TRUE FALSE

jmp cseg_84DD3

FALSE TRUE

mov ax, word ptr D7
mov word ptr D0, ax
shl word ptr D0, 5
call FindClosestPlayerToBallFacing
cmp A0, -1
jz cseg_84D10

FALSE

mov ax, word ptr D5
mov word ptr D0, ax
add word ptr D0, 16
and word ptr D0, 0FFh
shr word ptr D0, 5
mov ax, word ptr D7
sub word ptr D0, ax
and word ptr D0, 7
jz short cseg_84FA0

TRUE

mov esi, A5
mov ax, [esi+Sprite.teamNumber]
mov word ptr D0, ax
mov esi, A0
mov ax, [esi]
cmp word ptr D0, ax
jz @@our_player_closest

FALSE

mov esi, A6
mov ax, [esi+TeamGeneralInfo.field_84]
or ax, ax

jnz cseg_84DD3

TRUE

jmp cseg_84D10

FALSE

TRUE

cmp A0, -1
jnz @@our_player_closest

mov esi, A6
mov ax, [esi+130]
mov word ptr D0, ax
and word ptr D0, 3
jnz cseg_84DD3

FALSE

TRUE

jmp @@our_player_closest

FALSE

TRUE

cmp D6, 48400
ja cseg_84F4B

FALSE

mov ax, currentGameTick
mov word ptr D0, ax
and word ptr D0, 0Ch
jnz cseg_84DD3

TRUE

TRUE FALSE

mov ax, word ptr D7
mov word ptr D0, ax
shl word ptr D0, 5
call FindClosestPlayerToBallFacing
cmp A0, -1
jz short cseg_84D10

FALSE

TRUE

mov esi, A5
mov ax, [esi]
mov word ptr D0, ax
mov esi, A0
mov ax, [esi]
cmp word ptr D0, ax
jz @@our_player_closest

FALSE

TRUE

jmp short $+2

FALSE

TRUE

mov ax, frameCount
mov word ptr D0, ax
and word ptr D0, 7Fh
cmp word ptr D0, 20h
jnb cseg_84DD3

FALSE

TRUE

mov esi, A6
mov [esi+TeamGeneralInfo.field_84], 4

FALSE TRUE

test byte ptr currentGameTick, 80h; alternate sides each 128 frames
jz short @@turn_left

FALSE

mov esi, A5
mov ax, [esi+Sprite.direction]
mov esi, A6
mov [esi+TeamGeneralInfo.currentAllowedDirection], ax
retn

TRUE

mov esi, A5
mov ax, [esi+Sprite.direction]
mov word ptr D0, ax
add word ptr D0, 1
and word ptr D0, 7
mov ax, word ptr D0
mov esi, A6
mov [esi+TeamGeneralInfo.currentAllowedDirection], ax
retn

FALSE

mov esi, A5
mov ax, [esi+Sprite.direction]
mov word ptr D0, ax
sub word ptr D0, 1
and word ptr D0, 7
mov ax, word ptr D0
mov esi, A6
mov [esi+TeamGeneralInfo.currentAllowedDirection], ax
retn

TRUE

mov ax, word ptr D5
mov word ptr D0, ax
add word ptr D0, 16
and word ptr D0, 0FFh
shr word ptr D0, 5
mov ax, word ptr D0
mov esi, A6
mov [esi+TeamGeneralInfo.currentAllowedDirection], ax
retn

FALSE

TRUE

mov esi, A5
mov ax, [esi+Sprite.fullDirection]
mov word ptr D0, ax
add byte ptr D0, 128
add word ptr D0, 16
and word ptr D0, 0FFh
shr word ptr D0, 5; D0 = opposite player direction
mov esi, A5
mov ax, [esi+Sprite.direction]
cmp word ptr D0, ax
jz short @@set_opposite_direction

mov esi, A5
cmp [esi+Sprite.ballDistance], 800; 20^2 + 20^2
jb short @@set_opposite_direction

FALSE

mov ax, word ptr D0
mov esi, A6
mov [esi+TeamGeneralInfo.currentAllowedDirection], ax
or ax, ax
retn

TRUE

mov ax, currentGameTick; decide randomly whether to flip direction
mov word ptr D1, ax
and word ptr D1, 0Eh
jz short @@set_opposite_direction

FALSE

TRUE

mov esi, A5
mov ax, [esi+Sprite.direction]
mov esi, A6
mov [esi+TeamGeneralInfo.currentAllowedDirection], ax
retn

TRUE FALSE

mov ax, word ptr D7
mov esi, A6
mov [esi+TeamGeneralInfo.currentAllowedDirection], ax
or ax, ax
retn

call IsCpuReadyToResumePlay
jnb return; wait a little while for player to get rid of the ball

FALSE

TRUE

mov AI_resumePlayTimer, 15
mov ax, word ptr D7; D7 should be controlled player direction...
mov esi, A6
mov [esi+TeamGeneralInfo.currentAllowedDirection], ax
or ax, ax

mov esi, A6
mov [esi+TeamGeneralInfo.quickFire], 1; fire! (passing)
mov ax, AI_maxStoppageTime
mov word ptr D0, ax
mov ax, stoppageTimerActive
cmp word ptr D0, ax
ja short @@jmp_return

FALSE

TRUE mov ax, stoppageTimerActive
mov AI_maxStoppageTime, ax

FALSE

jmp return

TRUE

cmp word ptr D0, 1
jz short cseg_84FA0

FALSE

mov esi, A6
mov [esi+TeamGeneralInfo.AI_afterTouchStrength], 2
cmp D6, 115600
ja short cseg_84FCA

TRUE

cmp word ptr D0, -1
jz short cseg_84FA0

FALSE

TRUE

jmp cseg_84D10

TRUE FALSE

mov esi, A6
mov [esi+TeamGeneralInfo.AI_afterTouchStrength], 1

FALSE

mov ax, word ptr D7
mov esi, A6
mov [esi+TeamGeneralInfo.currentAllowedDirection], ax
mov esi, A6
mov [esi+TeamGeneralInfo.normalFire], 1
mov ax, word ptr D7
mov word ptr D2, ax
shl word ptr D2, 5
mov al, byte ptr D5
sub byte ptr D2, al
mov AI_resumePlayTimer, 15
mov word ptr D0, -1
mov al, byte ptr D2
or al, al
jns short cseg_85025

TRUE

neg word ptr D0

FALSE

mov eax, ballSprite.deltaY
or eax, eax

js short cseg_8503F

TRUE

cmp word ptr ballSprite.y+2, 555
ja cseg_850E1

FALSE

cmp word ptr ballSprite.y+2, 342
jb cseg_850E1

TRUE

jmp short cseg_8504E

FALSE

mov ax, word ptr D0
mov esi, A6
mov [esi+TeamGeneralInfo.AI_ballSpinDirection], ax
jmp return

TRUE

mov ax, currentGameTick
mov word ptr D1, ax
and word ptr D1, 1Ch
shr word ptr D1, 2
cmp word ptr ballSprite.x+2, 193
jb short cseg_85080

FALSE

TRUE

cmp word ptr ballSprite.x+2, 478
jb short cseg_85097

FALSE

cmp word ptr ballSprite.x+2, 118
jb short cseg_850C5

TRUE

FALSE

mov ax, word ptr D1
or ax, ax

jz short cseg_850CF

TRUE

cmp word ptr ballSprite.x+2, 553
ja short cseg_850C5

FALSE

cmp word ptr D1, 4
jb short cseg_850E1

TRUE

jmp short cseg_850AE

TRUE FALSE

mov ax, word ptr D1
or ax, ax

jz short cseg_850DA

cmp word ptr D1, 5
jb short cseg_850E1

FALSE

mov word ptr D0, 0
jmp short cseg_850E1

TRUE

jmp short cseg_850DA

FALSE

TRUE

neg word ptr D0

cmp word ptr D1, 4
jb short cseg_850CF

FALSE

TRUE

jmp short cseg_850E1

FALSE TRUE

FALSE

TRUE

call IsCpuReadyToResumePlay
jnb return

FALSE

TRUE

mov esi, A6
mov [esi+TeamGeneralInfo.AI_afterTouchStrength], 0
jmp short cseg_85178

FALSE

TRUE

mov AI_resumePlayTimer, 15
mov ax, word ptr D7
mov esi, A6
mov [esi+TeamGeneralInfo.currentAllowedDirection], ax
mov esi, A6
mov [esi+TeamGeneralInfo.normalFire], 1
mov word ptr D0, -1
mov al, byte ptr D2
or al, al
jns short cseg_851B4

mov ax, AI_resumePlayTimer
or ax, ax

jnz return

call IsCpuReadyToResumePlay
jnb return

FALSE

TRUE

mov esi, A6
mov [esi+TeamGeneralInfo.AI_afterTouchStrength], 1
jmp short cseg_85178

FALSE

TRUE

mov ax, AI_resumePlayTimer
or ax, ax

jnz return

call IsCpuReadyToResumePlay
jnb return

FALSE

TRUE

mov esi, A6
mov [esi+TeamGeneralInfo.AI_afterTouchStrength], 2

FALSE

TRUE

neg word ptr D0

FALSE

mov ax, word ptr D0
mov esi, A6
mov [esi+TeamGeneralInfo.AI_ballSpinDirection], ax
jmp return

TRUE

mov ax, stoppageTimerActive
mov AI_maxStoppageTime, ax

FALSE

mov AI_resumePlayTimer, 15
cmp gameState, ST_PENALTY
jz @@weak_after_touch

TRUE

cmp gameState, ST_PENALTIES
jz @@weak_after_touch

FALSE

mov esi, A6
mov [esi+TeamGeneralInfo.AI_afterTouchStrength], 0

TRUE

cmp gameState, ST_FREE_KICK_OUTER_LEFT
jb short @@test_corner

FALSE

TRUE

cmp gameState, ST_FREE_KICK_OUTER_RIGHT
jbe short @@check_distance_from_the_goal

FALSE

cmp gameState, ST_CORNER_LEFT
jz short @@medium_after_touch

TRUE

FALSE

cmp D6, 28800
jb short @@weak_after_touch

TRUE

nop
nop
nop
nop
cmp gameState, ST_CORNER_RIGHT
jz short @@medium_after_touch

FALSE

mov esi, A6
mov [esi+TeamGeneralInfo.AI_afterTouchStrength], 1
jmp short @@activate_normal_fire

TRUE

mov ax, AI_rand
mov word ptr D0, ax
and word ptr D0, 18h
jnz short @@check_distance_from_the_goal

FALSE

TRUE cmp word ptr D0, 10h
jz short @@weak_after_touch

TRUE FALSE

cmp word ptr D0, 8
jz short @@medium_after_touch

TRUE FALSE

jmp short @@strong_after_touch

FALSE TRUE

mov esi, A6
mov [esi+TeamGeneralInfo.AI_afterTouchStrength], 2
jmp short @@activate_normal_fire

cmp D6, 57800
jb short @@medium_after_touch

TRUE FALSE

FALSE

TRUE

mov ax, word ptr D7
mov esi, A6
mov [esi+TeamGeneralInfo.currentAllowedDirection], ax
mov esi, A6
mov [esi+TeamGeneralInfo.normalFire], 1
cmp gameState, ST_PENALTY
jz @@no_after_touch

cmp gameState, ST_PENALTIES
jz @@no_after_touch

FALSE

mov esi, A6
mov [esi+TeamGeneralInfo.AI_ballSpinDirection], 0
jmp return

TRUE

cmp gameState, ST_CORNER_LEFT
jz @@corner

FALSE

TRUE

cmp gameState, ST_CORNER_RIGHT
jz @@corner

FALSE

mov ax, AI_rand
mov word ptr D0, ax
or ax, ax

and word ptr D0, 7
cmp word ptr D0, 3
jb short cseg_853FA

TRUE

mov AI_resumePlayTimer, 15
mov ax, word ptr D7
mov esi, A6
mov [esi+TeamGeneralInfo.currentAllowedDirection], ax
mov esi, A6
mov [esi+TeamGeneralInfo.normalFire], 1
mov word ptr D0, 0
cmp gameState, ST_FOUL
jnz short cseg_8536B

TRUE FALSE

cmp A6, offset topTeamData
jnz short cseg_8535D

FALSE

mov word ptr D0, -1
mov al, byte ptr D2
or al, al
jns short cseg_85384

TRUE

mov esi, A2
cmp word ptr [esi+(Sprite.y+2)], 682
jge short cseg_85384

FALSE

mov esi, A2
cmp word ptr [esi+(Sprite.y+2)], 216
jle short cseg_85384

TRUE

jmp short cseg_8536B

FALSE

mov ax, word ptr D0
mov esi, A6
mov [esi+TeamGeneralInfo.AI_ballSpinDirection], ax
jmp return

TRUE

FALSE

TRUE

neg word ptr D0

FALSE

TRUE

cmp word ptr D0, 6
jb short cseg_853DB

FALSE

mov word ptr D1, 1
mov ax, word ptr D7
mov word ptr D0, ax
add word ptr D0, 1

TRUE

FALSE

mov word ptr D1, -1
mov ax, word ptr D7
mov word ptr D0, ax
sub word ptr D0, 1
jmp short cseg_85417

TRUE

and word ptr D0, 7
mov cl, byte ptr D0
mov ax, 1
shl ax, cl
test playerTurnFlags, ax
jz short @@no_after_touch

TRUE

mov ax, word ptr D1
mov esi, A6
mov [esi+TeamGeneralInfo.AI_ballSpinDirection], ax
jmp return

FALSE

cmp A4, 0; A4 -> pass to player sprite
jz short @@pass_to_player_too_far_or_null

FALSE

TRUE

mov esi, A4
mov eax, [esi+Sprite.ballDistance]
mov D0, eax
mov esi, A5
mov eax, [esi+Sprite.ballDistance]
sub D0, eax
cmp D0, 50; 5^2 + 5^2
jl short @@pass_to_player_too_far_or_null

FALSE

cmp A4, 0
jz @@decide_if_flipping_direction

TRUE

mov esi, A6; ball closer to the player that's receiving the pass
mov eax, [esi+TeamGeneralInfo.passToPlayerPtr]
mov D0, eax
mov esi, A6
mov eax, [esi+TeamGeneralInfo.controlledPlayer]
mov esi, A6
mov [esi+TeamGeneralInfo.passToPlayerPtr], eax
mov eax, D0
mov esi, A6
mov [esi+TeamGeneralInfo.controlledPlayer], eax
jmp return

FALSE TRUE

TRUE

mov ax, deadVarAlways0
or ax, ax

jnz cseg_84DE0

FALSE

TRUE

mov ax, topTeamData.playerNumber
or ax, ax

jnz short @@jmp_decide_if_flipping_direction

FALSE

mov ax, bottomTeamData.playerNumber
or ax, ax

jnz short @@jmp_decide_if_flipping_direction

FALSE

jmp @@decide_if_flipping_direction

TRUE

jmp short @@randomly_flip_or_continue_direction

FALSE TRUE

mov ax, currentGameTick
mov word ptr D0, ax
and word ptr D0, 18h; 12.5% chance of direction flip
jnz @@use_current_player_direction

mov esi, A4
mov eax, [esi+Sprite.ballDistance]
mov D0, eax
mov esi, A5
mov eax, [esi+Sprite.ballDistance]
sub D0, eax
cmp D0, 800
jg @@decide_if_flipping_direction

FALSE TRUE

TRUE

mov ax, AI_rand
mov word ptr D0, ax
and word ptr D0, 2
mov A0, offset AI_randomRotateTable
mov esi, A0
movzx ebx, word ptr D0
mov ax, [esi+ebx]
mov word ptr D0, ax
mov esi, A5
mov ax, [esi+Sprite.fullDirection]
add word ptr D0, ax; rotate left/right one position randomly
add byte ptr D0, 128; flip direction
add word ptr D0, 16
and word ptr D0, 0FFh
shr word ptr D0, 5
mov ax, word ptr D0
mov esi, A6
mov [esi+TeamGeneralInfo.currentAllowedDirection], ax
retn

FALSE

cmp A4, 0
jz @@decide_if_flipping_direction

TRUE

mov esi, A5
cmp [esi+Sprite.ballDistance], 2450
ja @@decide_if_flipping_direction

FALSE

TRUE

mov esi, A5
mov ax, [esi+Sprite.direction]
mov word ptr D0, ax
or ax, ax

js @@decide_if_flipping_direction

FALSE

TRUE

shl word ptr D0, 5
mov al, byte ptr D5
sub byte ptr D0, al
cmp byte ptr D0, 15
jg @@decide_if_flipping_direction

FALSE

TRUE cmp byte ptr D0, 241
jl @@decide_if_flipping_direction

FALSE

TRUE

mov ax, word ptr D7
mov esi, A6
mov [esi+TeamGeneralInfo.currentAllowedDirection], ax
mov esi, A6
mov [esi+TeamGeneralInfo.fireThisFrame], 1
jmp return

FALSE

mov ax, penalty
or ax, ax

jnz short @@no_ball_after_touch

FALSE

mov esi, A6
cmp [esi+TeamGeneralInfo.AI_afterTouchStrength], 1
jnz short @@do_long_kick

TRUE

mov ax, AI_rand
mov word ptr D0, ax
and word ptr D0, 1
jz short @@no_ball_after_touch

FALSE

TRUE

mov esi, A6
mov ax, [esi+TeamGeneralInfo.AI_ballSpinDirection]
or ax, ax

js short @@apply_left_spin

FALSE

TRUE

jnz short @@apply_right_spin

FALSE

mov ax, word ptr D1
mov word ptr D0, ax
sub word ptr D0, 1
and word ptr D0, 7
mov A0, offset AI_leftSpinTable
jmp short @@set_new_direction_for_kick_pass

TRUE

FALSE

mov ax, word ptr D1
mov word ptr D0, ax
add word ptr D0, 1
and word ptr D0, 7
mov A0, offset AI_rotateRightTable

TRUE

mov esi, A6
mov [esi+TeamGeneralInfo.currentAllowedDirection], -1
retn

FALSE

mov A0, offset AI_longKickTable
jmp short @@set_new_direction_for_kick_pass

TRUE

mov esi, A6
mov ax, [esi+TeamGeneralInfo.controlledPlDirection]
mov esi, A6
mov [esi+TeamGeneralInfo.currentAllowedDirection], ax
retn

mov esi, A6
mov ax, [esi+TeamGeneralInfo.AI_afterTouchStrength]
mov word ptr D0, ax
shl word ptr D0, 1
mov esi, A0
movzx ebx, word ptr D0
mov ax, [esi+ebx]
add word ptr D1, ax
and word ptr D1, 7
mov ax, word ptr D1
mov esi, A6
mov [esi+TeamGeneralInfo.currentAllowedDirection], ax
retn